Why Assumptions & Uncertainties in Bed DEM's Matter: A Geothermal Example from Haynes Glacier, West Antarctica

Jason Bott[1], Shuai Yan[2], Don Blankenship[1], Duncan Young[1]
[1]University of Texas, Institute for Geophysics, Austin, United States
[2]Department of Earth Sciences, University of Washington, Seattle, United States

Email: jbott@utexas.edu

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Introduction

- **Bed Topography** is *the* most important boundary condition for ice modeling
- **Circular logic** and **large inaccuracies** exist in widely used realizations of subglacial topography (Bedmap, BedMachine, etc.)
- **Stochastic Simulation** and other statistical techniques could improve the accuracy of ice models

Why Haynes Glacier?

- **Trough-Dominated** Basal Morphology → Bedmap's Weakness
- **Heterogenous & Anomalous Geothermal Flux** likely from **Active Volcanism** → BedMachine's Weakness
- Rich array of **Bright Radar Horizons** (Possibly Ash Layers) which help constrain internal ice dynamics

Where Are We?



1

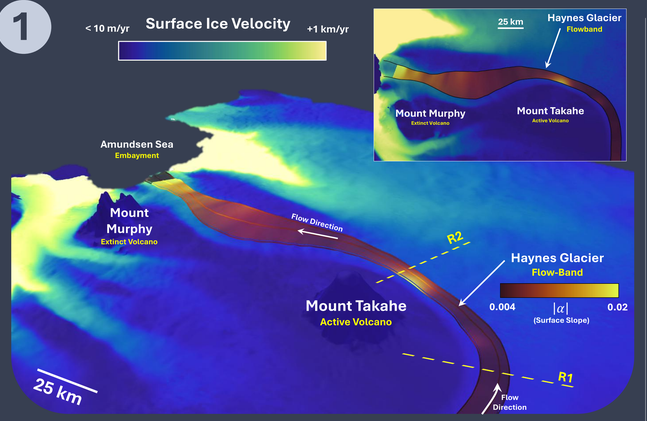


Figure 1: Haynes Glacier flow-band traced along the INSAR-based MEaSUREs v2 surface ice velocity dataset. Surface slope was calculated along the flow-band and is shown above. The flow-band's width helps constrain convergent/divergent flow along the path of ice flow.

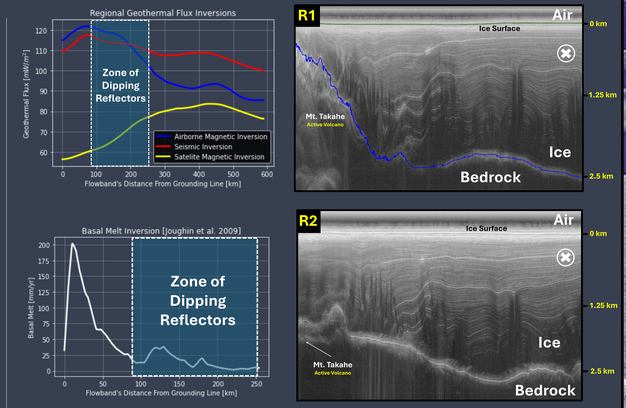
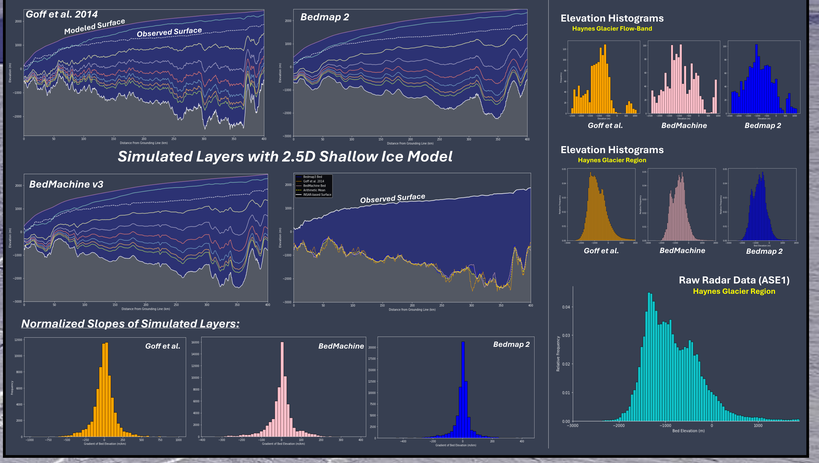


Figure 2: Evidence for high-magnitude and spatially variable geothermal flux at Haynes Glacier, West Antarctica. Dipping englacial reflectors serve as a likely indicator of enhanced basal melt from anomalous geothermal flux. Comparison to numerically modeled basal friction from *Joughin et al. 2009* helps constrain the contribution of heat from basal friction.

2

$$q(x) = W(x)\cdot\int_z^H u(x_0,z)dz + \int_{x_0}^x \dot{b}(x) - \dot{m}(x)dx = W(x)\cdot\frac{2A}{n+2}(\rho g\alpha)^n H^{n+2} - \int_{x_0}^x \dot{m}(x)dx$$

Total Mass Flux = Kinematic Mass Flux + Accumulation – Basal Melt = Shallow Ice Approx. of Mass Flux - Basal Melt

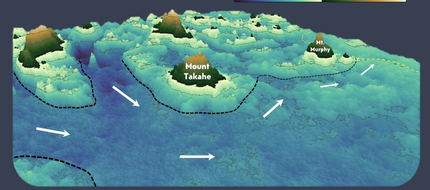


3

Future Work:

1. Reconciliation the elevation and geometry of modeled vs. observed layers
2. Propagation of stochastic elements into forward model
3. Construction of improved metrics for the forward model's sensitivity to bed DEMs

Goff et al. 2014



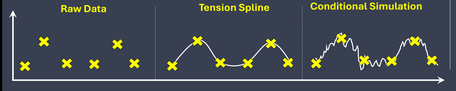
$$Z_C(\vec{x}) = Z_I(\vec{x}) + Z_U(\vec{x}) - Z_{UI}(\vec{x})$$

Deterministic Interpolation + Stochastic Roughness − Interpolation of Stochastic Sim.

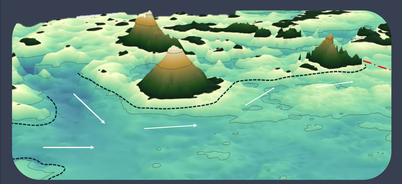
1) $Z_I(\vec{x}) = f(\vec{x}) = \sum_{j=1}^{N} c_j\phi(\vec{x}-\vec{x_j})$ *subject to:* $\varepsilon\nabla^4 f - \tau\nabla^2 f = 0$

Smooths by adding adjustable "rigidity"

2) $Z_C(\vec{x}) = C_{hh}(\vec{x}) = H^2 G_v \frac{r(\vec{x})}{G_v(0)}$



BedMachine v3



$$\begin{cases} \nabla\cdot H\bar{v} = \dot{a}\,, & in\ \Omega \\ H = H_{obs}, & on\ \Gamma \end{cases}$$

General Mass-Conservation Model Setup

1) $\min_H \left[\int_T \frac{1}{2}(H-H_{obs})^2 dT + \int_\Omega \gamma_1(\nabla H\cdot\vec{n}_1)^2\, d\Omega + \int_\Omega \gamma_\perp(\nabla H\cdot\vec{n}_\perp)^2\, d\Omega\right]$

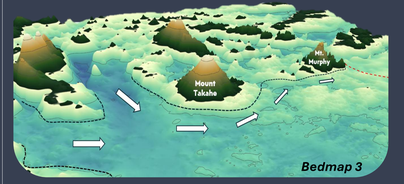
Fits Modeled Thickness to Observed Data

Regularization of ice thickness along and across flow directions respectively

2) $H(\vec{x}) = \frac{\sum_{i=1}^N \omega_i(\vec{x})H_i}{\sum_{i=1}^N \omega_i(\vec{x})}$ ← $\begin{cases} \omega_i(\vec{x}) = \frac{1}{\|\vec{x}-\vec{x}_i\|^n} \\ n = 2 \end{cases}$

Inverse Distance Weighting to Create an Interpolated Data Product

Bedmap 2 & 3

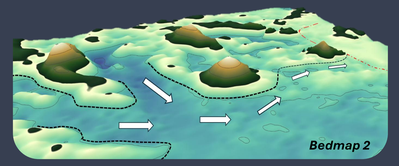


$$f(\vec{x}) = a_0 + a_1x_1 + a_2x_2 + \sum_{i=1}^{N}\omega_i\phi\|\vec{x}-x_i\|_2$$

Interpolated function informed by local statistics

Subject to: $\min_f \iint \|\nabla^2 f\|^2 dxdy$

Curvature Minimization (Smoothing)



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References